RISHABH SHARMA

CURRICULUM VITAE

CONTACT

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ABOUT ME

I am passionate about mathematics and problem solving. My major interests are in Number theory, Geometry and Algebra.

My hobbies include climbing and solving puzzles.
I am also into Indian Classical Music and enjoy playing the flute.

LANGUAGES

- English
  Professional Working Proficiency
- Hindi
  Native
- German
  Elementary Proficiency
- Dutch
  Elementary Proficiency
- French
  Elementary Proficiency

SKILLS

- Python, C++, C, Haskell, JAVA
- LaTeX

EDUCATION

ALGANT (LEIDEN & REGENSBURG UNIVERSITY) — *Leiden, The Netherlands*
MSc. program in mathematics — *2024-2025*
with 1st year at Leiden university — *Regensburg, Germany*
and the 2nd year at Regensburg university — *2025-2026 (Expected)*

CHENNAI MATHEMATICAL INSTITUTE — Chennai, India
BSc. Honors — *2021 - 2024*
Mathematics and Computer Science

DELHI PUBLIC SCHOOL, FARIDABAD — Haryana, India
CBSE Grade 12 — *2019 - 2021*
97.4%

DELHI PUBLIC SCHOOL, FARIDABAD — Haryana, India
CBSE Grade 10 — *2007 - 2018*
97.2%

COURSES TAKEN (CLICK HERE TO SEE COURSE INFORMATION)

- Real analysis on R, R^n, metric spaces and function spaces.
- Multivariable Differential and Integral Calculus.
- Linear Algebra, Group Theory, Ring Theory, Field and Galois theory.
- Probability, Discrete Mathematics.
- Topology, Differential Equations, Complex analysis, Differential Geometry
- Haskell, Advanced Programming with Python, Theory of Computation, Algorithms, Programming Language Concepts.
- Elementary Number theory, Algebraic Number theory, Local Fields
- Analytic Number theory, Distribution of Primes
- Arithmetic of Elliptic curves, Varieties and Scheme theory
- Commutative Algebra
- Differential Geometry, Algebraic Topology
- Stochastic Processes, Twisted (co-)homology and Reidemeister Torsion

PROJECTS AND EXPERIENCE

MASTER'S THESIS April 2026 - August 2026 (expected)

On the Explicit Class Field theory of CM fields. Constructing abelian extensions of CM fields via points on CM abelian varieties.
A draft of a chapter can be found here.

COUNSELLOR AT PROMYS BOSTON June 2025 - August 2025

Worked at the parent program of PROMYS India held at Boston University. In addition to giving feedback and guiding students on Number theory, I also gave counsellor seminars and helped in running the advanced seminar on Quivers and their representations given by Prof. Brian Williams.

COUNSELLOR AT PROMYS INDIA May 2024 - June 2024

I worked with and mentored several students at this prestigious math camp for high school students held at IISc Bangalore. Duties involved giving feedback on daily problem sets for 3 students and subsequent problem discussions.

TEACHING ASSISTANT January 2024 - April 2024

I was a teaching assistant for the course Algebra 1 (Linear Algebra) instructed by Prof. Upendra Kulkarni at CMI.
This couse is aimed at 1st year undergraduates.

READING PROJECT September 2023 - November 2023

Advisor: Prof. Sushmita Venugopalan (IMSc)
Topic: Algebraic curves
I read and worked through the book "Algebraic curves: Towards Moduli Spaces" by Lando and Prasolov, under the guidance of Prof. Venugopalan.

READING PROJECT July 2023 - November 2023

Advisor: Prof. Siddhi Pathak (CMI)
Topic: Modular forms
I read and worked through the book "Problems in the Theory of Modular forms" by Ram Murty, under the guidance of Prof. Pathak. I also read a paper on the use of modular forms in irrationality proofs, in particular, the irrationality of ζ(2) and ζ(3). The paper is available here.

READING PROJECT February 2023 - August 2023

Advisor: Prof. Aditya Karnataki (CMI)
Topic: Algebraic Number theory
I read and worked through the books "Number Fields" by Daniel Marcus and "A classical introduction to modern number theory" by Ireland & Rosen under the guidance of Prof. Karnataki. Also worked on a version of the inverse galois problem for small groups. A report is available here.

STEMS MATHS Team Co-Head October 2022 - February 2023

STEMS (Scholastic Test of Excellence in Mathematical Sciences) is a nationwide open-book exam conducted by the students of CMI every year as part of the college fest of CMI, Tessellate.

Our team created, collected, reviewed and proposed problems for the math exam in STEMS.

READING PROJECT December 2022

Advisor: Prof. R. Balasubramanian (IMSc)
Topic: Algebraic Number theory

TEACHING ASSISTANT August 2022 - November 2022

I was a teaching assistant for the course Analysis 2 (Multivariable Differential Calculus) instructed by Prof. Upendra Kulkarni at CMI.
This course is aimed at 1st year undergraduates

ACHIEVEMENTS

Shriram Scholarship Awardee (CMI) *2021*
A 100% waiver on tuition in CMI
Awarded by CMI and the Shriram Group

JEE ADVANCED Qualified *2021*
All India Rank - 624

INMO Awardee & IMOTC Attendee *2020*
The Indian National Mathematical Olympiad is a selection stage for the Indian team to represent India at the prestigious IMO.
IMOTC is the training camp for preparation towards the IMO.

KVPY Scholar *2019*
Kishore Vaigyanik Protsahan Yojana conducted by IISc, Bangalore.
All India Rank - 797

NTSE Scholar *2019*
National Talent Search Examination conducted by CBSE